

Power BI Interview Question Bank

A Complete Guide — Basic to Advanced



How to Use This Question Bank

This bank is organized into **5 levels**:

1. **Basics / Fundamentals** — for freshers or 0–1 year experience
2. **Intermediate** — data modeling, DAX fundamentals, visuals
3. **Advanced** — complex DAX, performance, security
4. **Scenario-Based / Real-World** — problem-solving and judgment
5. **Behavioral & Project-Based** — for gauging real hands-on experience

Each question includes a **brief answer/pointer** so recruiters (even non-technical ones) can evaluate responses, and candidates can use it for self-prep.

LEVEL 1: BASICS / FUNDAMENTALS

1. What is Power BI?

A business analytics tool by Microsoft used to visualize data, build interactive dashboards/reports, and share insights across an organization.

2. What are the components of Power BI?

- **Power BI Desktop** – report/dashboard authoring tool
- **Power BI Service** – cloud-based platform for sharing/collaboration
- **Power BI Mobile** – app for viewing reports on mobile devices
- **Power BI Report Server** – on-premise reporting
- **Power BI Gateway** – connects on-premise data to the cloud
- **Power Query** – data transformation engine
- **Power Pivot** – in-memory data modeling engine

3. What is the difference between Power BI Desktop and Power BI Service?

Desktop is used to build reports/dashboards locally; Service is the cloud platform used to publish, share, schedule refreshes, and collaborate.

4. What file formats does Power BI use?

- .pbix – Power BI Desktop file (report + data + model)
- .pbit – Power BI template file (structure without data)
- .pbids – Power BI data source file

5. What is Power Query?

A data connection and transformation tool inside Power BI (and Excel) used for ETL (Extract, Transform, Load) using the M language.

6. What is DAX?

Data Analysis Expressions — a formula language used to create calculated columns, measures, and custom calculations in Power BI.

7. What are the different types of data connectivity modes in Power BI?

- **Import** – data is loaded into Power BI's in-memory engine
- **DirectQuery** – queries live data from the source, no data stored locally
- **Live Connection** – connects directly to Analysis Services/Power BI datasets
- **Composite/Dual mode** – mix of Import and DirectQuery

8. What is the difference between a calculated column and a measure?

A calculated column is computed row-by-row and stored in the table (uses memory); a measure is calculated at query/render time based on filter context (dynamic, more memory-efficient).

9. What types of visuals are available in Power BI?

Bar/column charts, line charts, pie/donut charts, maps, tables, matrices, cards, gauges, KPIs, slicers, treemaps, scatter plots, and custom visuals from the marketplace.

10. What is a slicer?

A visual filter control that lets users interactively filter report data by a chosen field (e.g., Region, Date, Category).

11. What is the difference between a filter and a slicer?

A slicer is a visible, on-canvas interactive control; a filter (in the Filters pane) can be applied at visual, page, or report level and isn't always visible on the canvas.

12. What is Power BI Gateway used for?

It enables secure data transfer between on-premise data sources and the Power BI cloud service for scheduled refreshes.

13. What are the types of gateways?

- **Personal Gateway** – single user, no sharing
- **Standard/Enterprise Gateway** – can be shared across an organization, supports multiple data sources

14. What is a dashboard vs. a report in Power BI?

A report is a multi-page collection of visuals built from one dataset with interactivity/filtering; a dashboard is a single-page, high-level canvas pinned from one or more reports, primarily for monitoring.

15. Can Power BI connect to Excel?

Yes, Power BI can import Excel files directly or connect live via Excel's data model.

LEVEL 2: INTERMEDIATE

16. What is data modeling in Power BI?

The process of defining relationships between tables (fact and dimension tables) to enable accurate cross-table analysis, typically following a star schema.

17. What is a Star Schema vs Snowflake Schema?

- **Star schema:** a central fact table connected directly to denormalized dimension tables (simpler, faster, preferred in Power BI)
- **Snowflake schema:** dimension tables are further normalized into sub-dimensions (more complex, more joins)

18. What are Fact and Dimension tables?

- **Fact table:** contains measurable, quantitative data (sales amount, quantity)
- **Dimension table:** contains descriptive attributes (product name, customer, date)

19. What are relationships in Power BI, and what types exist?

Links between tables based on common columns.

- One-to-Many (most common)
- One-to-One
- Many-to-Many

20. What is cardinality in Power BI?

It defines how many matching rows exist on each side of a relationship between two tables (e.g., one-to-many, many-to-many).

21. What is cross-filter direction?

It determines whether filters flow in a single direction or both directions between related tables — **Single** or **Both**.

22. Explain context in DAX — Row Context vs Filter Context.

- **Row context:** refers to the "current row" during row-by-row evaluation (e.g., in calculated columns)
- **Filter context:** the set of filters applied to a calculation based on slicers, visuals, or filters, affecting the result of a measure

23. What is the CALCULATE function in DAX?

One of the most powerful DAX functions — it evaluates an expression within a modified filter context, allowing you to override or add filters.

24. Explain some common DAX functions.

- SUM(), AVERAGE(), COUNT(), COUNTROWS()
- IF(), SWITCH()
- RELATED(), RELATEDTABLE()
- FILTER(), ALL(), ALLEXCEPT()
- SUMX(), AVERAGEX() (iterator functions)

25. What is the difference between SUM() and SUMX()?

SUM() aggregates a single column directly; SUMX() is an iterator that evaluates an expression row-by-row across a table and then sums the result — used for calculations involving multiple columns (e.g., Price × Quantity).

26. What are Quick Measures?

Pre-built, commonly used DAX calculations (like running totals, percentage of total) that Power BI generates automatically through a UI, without writing DAX manually.

27. What is Power Query M language?

The functional programming language used behind the scenes in Power Query for data transformation steps (each applied step is generated in M).

28. What are some common data transformation steps in Power Query?

Removing duplicates, changing data types, splitting/merging columns, pivoting/unpivoting, merging queries (joins), appending queries (union), filtering rows, grouping data.

29. What is the difference between Merge Queries and Append Queries?

- **Merge** – joins two tables side-by-side based on a common column (like SQL JOIN)
- **Append** – stacks tables on top of each other (like SQL UNION), combining rows

30. What is a Hierarchy in Power BI?

A structured grouping of related fields (e.g., Year > Quarter > Month > Day) that allows drill-down/drill-up analysis in visuals.

31. What are bookmarks in Power BI?

Saved snapshots of a report's current state (filters, visual visibility, slicer selections) used to build interactive navigation or storytelling.

32. What is drill-through in Power BI?

A feature that lets users right-click a data point to navigate to a detailed report page filtered to that specific context.

33. What are KPIs in Power BI?

Visuals used to track a metric's progress against a defined target/goal, often showing trend indicators.

34. What is Row-Level Security (RLS)?

A feature that restricts data visibility for specific users based on roles, so different users see only the data relevant to them within the same report.

35. How do you publish a report to Power BI Service?

Via the "Publish" button in Power BI Desktop, which pushes the .pbix file (dataset + report) to a selected workspace in the Power BI Service.

LEVEL 3: ADVANCED

36. Explain the difference between ALL(), ALLEXCEPT(), and ALLSELECTED().

- ALL() – removes all filters from a table/column
- ALLEXCEPT() – removes all filters except those specified
- ALLSELECTED() – removes filters from inside the visual but retains filters from outside (e.g., slicers, page filters)

37. What is context transition in DAX?

When a row context is converted into an equivalent filter context — this happens automatically inside CALCULATE() or when using measures inside calculated columns.

38. What is the difference between DirectQuery and Import mode? When would you choose one over the other?

Import loads data into memory for fast performance but requires scheduled refresh and has data size limits; DirectQuery queries the live source in real time (no refresh needed, always current) but is slower and limited by source performance. DirectQuery is preferred for very large or real-time datasets; Import is preferred for performance and complex DAX.

39. What is a Composite Model?

A model combining Import and DirectQuery tables in the same report, enabling flexibility between performance and real-time data needs.

40. What are Aggregations in Power BI (Import/DirectQuery hybrid)?

A performance optimization technique where a summarized (aggregated) table is used for common queries, and DirectQuery is used only for detailed drill-through — significantly speeding up large datasets.

41. What is Query Folding in Power Query?

The process where Power Query translates transformation steps into a single native query executed at the data source, rather than pulling all raw data and transforming it locally — critical for performance.

42. How do you optimize a slow Power BI report?

- Reduce number of visuals per page
- Use aggregation tables
- Optimize DAX (avoid unnecessary iterators, avoid calculated columns where measures work)
- Disable unnecessary interactions between visuals
- Use star schema over snowflake
- Reduce cardinality of columns
- Use variables (VAR) in DAX to avoid recomputation
- Enable query folding
- Use Performance Analyzer to identify bottlenecks

43. What is the Performance Analyzer in Power BI?

A built-in tool that records and displays the time taken by each visual, DAX query, and rendering step, helping identify performance bottlenecks.

44. What is VertiPaq Engine?

Power BI's in-memory columnar storage engine (based on xVelocity) that compresses and stores imported data for fast query performance.

45. Explain VAR in DAX and why it's useful.

VAR allows you to store an intermediate calculation result within a DAX formula, improving readability and performance by avoiding repeated evaluation of the same expression.

46. What is the difference between a calculated table and a regular table?

A calculated table is generated using a DAX expression within the model (e.g., using SUMMARIZE, CALENDAR) rather than being imported from an external source.

47. What are Time Intelligence functions in DAX? Give examples.

Functions designed to work with dates for period-based comparisons: TOTALYTD(), TOTALQTD(), SAMEPERIODLASTYEAR(), DATEADD(), PARALLELPERIOD(), PREVIOUSMONTH(), DATESYTD()

48. What is a Date table, and why is it important?

A dedicated table containing continuous dates, marked as a "Date Table" in the model, required for Time Intelligence functions and consistent date-based filtering across fact tables.

49. What is the difference between Many-to-Many relationships and how do you handle them?

When both related columns contain duplicate values on each side, requiring careful handling (often via a bridge table) to avoid ambiguous or incorrect aggregation.

50. What is Incremental Refresh, and why is it used?

A feature that only refreshes new/changed data (instead of the whole dataset) based on defined date range policies — improves refresh performance for large datasets.

51. What is a Paginated Report (Power BI Report Builder)?

A pixel-perfect, print-ready report format (based on RDL/SSRS) used for highly formatted documents like invoices, unlike the interactive Power BI report canvas.

52. What is Dataflows in Power BI?

A cloud-based ETL tool that allows you to create reusable data preparation logic (using Power Query Online) stored in Azure Data Lake, shareable across multiple reports/datasets.

53. What is the XMLA endpoint, and what is it used for?

An interface that allows external tools (like SSMS, Tabular Editor, DAX Studio) to connect to Power BI datasets for advanced management, deployment, and querying — available in Premium/Premium Per User workspaces.

54. What is Tabular Editor / DAX Studio, and why would you use them?

Third-party tools used for advanced model editing, DAX query performance analysis, and bulk metadata management beyond what Power BI Desktop's UI supports.

55. Explain Object-Level Security (OLS) vs Row-Level Security (RLS).

RLS restricts specific *rows* of data for users/roles; OLS restricts access to entire *tables or columns*, hiding them completely from certain users.

56. What is a Shared Dataset vs a Report-level dataset?

A shared dataset is published independently in a workspace and can be reused across multiple reports; a report-level dataset is embedded within a single .pbix file.

57. How does Power BI handle version control / CI-CD?

Through Power BI Deployment Pipelines (Dev/Test/Prod stages), integration with Azure DevOps/Git for .pbip (Power BI Project) files, and external tools for automated deployment.

58. What is the difference between .pbix and .pbip file formats?

.pbix is a single compressed binary file; .pbip (Power BI Project) is a newer format that saves reports/datasets as individual text-based files, enabling proper Git version control and collaboration.

LEVEL 4: SCENARIO-BASED / REAL-WORLD QUESTIONS

59. "Your report is taking too long to load. Walk me through how you'd troubleshoot it."

Expected approach: use Performance Analyzer → check DAX query times vs. visual rendering times → review data model (star schema, cardinality) → check for unnecessary calculated columns → consider aggregations/incremental refresh → optimize DAX with variables.

60. "A client wants different sales managers to only see their own region's data. How would you implement this?"

Implement Row-Level Security (RLS) using a role based on a User/Region mapping table, using USERPRINCIPALNAME() and CALCULATE/FILTER logic.

61. "You need to combine data from SQL Server, Excel, and an API into one report. How do you approach this?"

Use Power Query to connect to each source individually, apply necessary transformations, ensure consistent schema/data types, then merge/append as needed in the data model, ideally following a star schema.

62. "The client wants the dashboard to always reflect real-time data. What would you recommend?"

Discuss DirectQuery vs. scheduled Import refresh vs. hybrid/composite models, and the performance trade-offs of each, recommending based on data volume and real-time need.

63. "How would you design a sales dashboard for a retail company from scratch?"

Expected answer covers: requirement gathering, identifying KPIs (revenue, margin, YoY growth), designing star schema (Sales fact + Product/Customer/Date/Store dimensions), building relevant DAX measures, and choosing appropriate visuals (trend lines, maps, KPIs).

64. "Two tables need a many-to-many relationship, but Power BI is giving ambiguous results. How do you fix it?"

Suggest creating a bridge/junction table to normalize the relationship into two one-to-many relationships instead of a direct many-to-many.

65. "The dataset has 50 million rows and refresh is timing out. What's your plan?"

Discuss incremental refresh, aggregation tables, reducing granularity, removing unused columns, query folding, and considering DirectQuery/composite models or Premium capacity.

LEVEL 5: BEHAVIORAL & PROJECT-BASED QUESTIONS

66. Walk me through a Power BI project you built end-to-end.

Look for: clear articulation of business problem → data sources → transformation logic → data modeling decisions → DAX measures created → visuals chosen and why → impact/result delivered.

67. Describe a time you had to optimize a poorly performing report.

Look for specifics: what tool they used (Performance Analyzer, DAX Studio), what the root cause was, and the measurable improvement.

68. How do you handle conflicting requirements from different stakeholders on the same dashboard?

Look for communication and prioritization skills — not just technical answers.

69. Have you worked with large datasets? How did you ensure performance stayed acceptable?

Should reference real techniques: aggregations, star schema design, incremental refresh, reducing column cardinality.

70. How do you ensure data accuracy in your reports?

Look for: validation against source systems, reconciliation checks, testing measures against known totals, peer review process.

71. How do you stay updated with new Power BI features?

Look for engagement with Microsoft's monthly Power BI updates, community blogs, Microsoft Learn, or certifications (PL-300).

72. Tell me about a time a report you built was misunderstood or misused by stakeholders. What did you do?

Assesses communication skills and ability to design intuitive, well-labeled dashboards with proper documentation/tooltips.

BONUS: Quick-Fire Rapid Screening Questions

Useful for a fast initial technical screen (10–15 min):

1. What's the shortcut/method to create a measure vs. a calculated column?
 2. Name three DAX time intelligence functions.
 3. What's the maximum file size for a .pbix file in Power BI Pro vs Premium?
 4. What's the difference between COUNT() and COUNTA()?
 5. Can Power BI connect to Python/R scripts? (Yes — for data transformation and custom visuals)
 6. What license type is needed for Row-Level Security to work in shared workspaces? (Pro or Premium)
 7. What's the refresh limit per day for Power BI Pro? (8 refreshes/day for Pro; 48/day for Premium)
 8. Name two ways to share a Power BI report with external users.
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Suggested Evaluation Rubric for Recruiters

Level	What to Listen For
Basic	Correct terminology, understands the BI workflow (connect → transform → model → visualize → share)
Intermediate	Comfortable with DAX fundamentals, understands data modeling and star schema, has built real reports
Advanced	Can discuss performance optimization, understands engine internals (VertiPaq), handles enterprise-scale concerns (RLS, incremental refresh, deployment pipelines)
Scenario-based	Structured problem-solving, asks clarifying questions, considers trade-offs rather than giving one-size-fits-all answers
Behavioral	Can clearly narrate real project experience with specifics, not just textbook definitions

This question bank can be adapted based on the seniority of the role — e.g., a Power BI Analyst role may stop around Level 2–3, while a Power BI Developer/Architect role should be tested through Level 4–5.